

Project overview and goals

For my project I analyzed a dataset regarding flights in the U.S. with the goal of creating a ML model that can predict whether a flight will be delayed, I also created a report in Looker Studio, allowing people to visualize the data I explored and analyze it themselves at a surface level using some embedded filters.

Link to my report: <https://datastudio.google.com/s/ildmZ3s7SOA>

Dataset description

My dataset was from the Bureau of Transportation Statistics (<https://www.bts.gov/>) specifically a dataset from the aviation category titled “Airline On-Time Performance Data”, my data covered the time range 2002-2025 and was roughly 38 GB in size with 150 million rows.

Some related dataset facts:

- 49 total attributes, 22 were integers, 15 were strings, 13 were floats
- The BTS (Bureau of Transportation Statistics) stores data about each month + year in separate files.
- Meaning my dataset came in 276 separate files which had to be downloaded.
- I used a script to download (data_downloader.py) and merge my data (merge_csv.py) together.

Description of pipeline workflow and services used (the overall architecture).

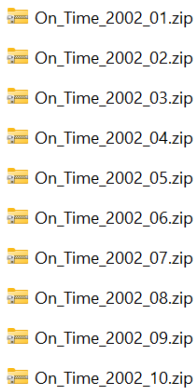
- 1) Use a script to slowly download all files from BTS website
- 2) Merge files together into one large CSV
- 3) Upload large CSV into a **Google Cloud Bucket**
- 4) From a bucket load data into **BigQuery**
- 5) Make a report in **Looker Studio** to explore my data, with the goal of understanding what impacts delays, while also making a nice dashboard.
- 6) Engage in model creation and evaluation in BigQuery, with the goal of predicting whether a flight will be delayed by 15 minutes or more. (Label attribute was “dep_del15)

Data processing steps: cleaning, transformations, and intermediate outputs (screenshots if helpful).

My data came in a clean state from the government website. There were no super obvious errors or mistakes I had to resolve before loading it into a bucket and then into BigQuery.

The real issue, like I mentioned briefly above, was that my data came in many different files, which if had wanted to download manually would have necessitated a lot of tedious work, filtering down to the correct one month + year period (The BTS only lets you download data in one month chunks at a time, after each chunk you have to change the filter settings to download another chunk).

Example of how files come from the BTS website:



On_Time_2002_01.zip
On_Time_2002_02.zip
On_Time_2002_03.zip
On_Time_2002_04.zip
On_Time_2002_05.zip
On_Time_2002_06.zip
On_Time_2002_07.zip
On_Time_2002_08.zip
On_Time_2002_09.zip
On_Time_2002_10.zip

To circumvent this, I used a script that sent requests to the website to get my files. After which I used another script to merge them all together into one large CSV file.

Later, I added 4 calculated fields to my dataset in order to improve the performance of my logistic regression classifier.

These transformations/calculations did things like:

- Change the departure time attribute from a format that combined hours and minutes together (like hhmm) into a range from 0 – 24.
- Found the flight frequency of specific routes
- Calculated the average delay rate of each departure airport.
- Calculated the average delay route of each specific route

I feel like some these calculations kind of toe the line of leaking data to the ML model about the outcome, but I feel like they warranted being included because in the real world routes are flown many hundreds and thousands of times over the years, and historical performance of flights (and their average delay rate), would be something companies have access to and can use to predict future performance.

Results obtained: what tables, summaries, or processed data came out of the pipeline

My project culminated in a ML classification model with a relatively good performance at predicting if a flight will be delayed by 15+ minutes, which I think is probably the largest tangible result.

```
1 SELECT *
2 FROM ML.EVALUATE(
3 | MODEL `flight_delay_ML.dep_delay15_logistic_v16`
4 );
```

✓ Query completed

Query results

Job information	Results	Visualization	JSON	Execution details	Execution graph	
Row	precision ▾	recall ▾	accuracy ▾	f1_score ▾	log_loss ▾	roc_auc ▾
1	0.793251554665...	0.719731870696...	0.914489095670...	0.754705459163...	0.234561568041...	0.908120879120...

In addition, the project led to a nice-looking dashboard people can use to explore the dataset and an understanding of some factors that have a large impact on flight delays.

Visualizations / dashboard

I made my visualization in looker studio, it's linked at the top of the report, but if you don't want to scroll back up you can find it here: <https://datastudio.google.com/s/ildmZ3s7SOA>

Overall, I am pretty satisfied with my visuals, I think they do a good job of illustrating the basics of the dataset and the dashboard provides (basic) tools for users to do their own data analysis if they are also interested in this topic.

Challenges encountered and how they were resolved

- 1) My data came in a very fragmented format from my data source, how I handled that was already mentioned above.
- 2) When I loaded my dataset into BigQuery from my bucket, I ran into errors twice for different reasons. The first came from BigQuery's auto generated schema running into issues as the datatype of a column switched (there were non-integer characters) a couple million rows in, this was resolved by manually creating a schema.
- 3) The second issue was just getting unspecified errors a couple of times when loading the table into BigQuery causing failure. As I didn't have an easy way of diagnosing the issue I just raised the number of acceptable errors when loading the data in from 0 to 5, which resolved the issue but likely slightly worsened data integrity/quality.

4) My initial model also had relatively low performance initially as the attribute I was attempting to classify had a pretty high-class imbalance (80% non-delayed, 20% delayed). I attempted to resolve this and boost performance by adding some calculated fields to the dataset I trained my model on which worked well.

Lessons learned / overview of what you learned

The dataset I used for this project was around ~38 GB, part of that was just the nature of the data, there are many flights every day, another reason is I chose a long time period lasting 23 years (2002-2025), another large reason is I opted to include a wide variety of attributes in the data I pulled from the BTS website. This last reason for my file size I want to focus on as a point of learning. The BTS website I downloaded the data from made every attribute an optional download, you could choose whether to download an attribute or skip over it. When I was starting out this project I began by casting a wide net and picking out 49 attributes I thought could be relevant to my project, I then proceeded to download them for large data range of 23 years. I did not use my full attribute list in my model, nor did I use it in my dashboard, in fact many of the attributes I initially downloaded for this project went unused when I finalized/narrowed my topic from airflight delays in general to specifically departure delays.

I think a better alternative approach to what I did would have been:

- 1) Load 49 attributes worth of data in for a small-time range (2002 January – 2002 June) into BigQuery and explore the data while determining what I want to use my model to predict.
- 2) Finalize an attribute to use as my label/target attribute and then make a list of all the relevant attributes somewhat related to what I want to predict.
- 3) Then download the full-time range of data for this much more limited attribute list.

Following the above process would have made my dataset much smaller, and subsequently the downloading and uploading of my dataset much faster. In the future if I'm engaging in projects like this, I intend to always do a cursory initial exploration of my data to narrow my attribute list prior to downloading my data in bulk (when possible). As opposed to downloading all my data followed by picking out the necessary attributes.

Additional things I learned more relevant to my specific project: The airport you depart from, the time of day, and the airline you fly with have a noticeable impact on your odds of receiving a delay. With flights later at night tending to have more delays. Additionally, for a given flight route (Departure->Arrival) the odds of receiving a delay have remained somewhat stable over the time period I tracked, meaning a routes historical delay rate (number of delays / total flights), can be a useful feature in a ML model.

Potential next steps if more time were available

Some obvious ideas are:

- 1) Add data about the weather and temperature at the departure airport to my model.
- 2) Find a way to include information about the planes model (both specific type, and general size)
- 3) If possible, see if delays are correlated with the captains who fly the planes (this is probably hard information to find?)
- 4) Add information about visibility, which would be a combination of time of day + time zone + weather. My thoughts being that the worse the visibility the slower things will move.
- 5) With an abundance of time (and patience), I would try creating a model using a “Boosted Tree”.
- 6) Investigate how the number of flights from a given airport (and a specific flight ranking in that order) impacts delays. (The 600th flight out of an airport probably is more prone to delays than the 20th, busy airports in general might have more delays).
- 7) Investigate how airport size/staffing levels impact delays, do small airports with fewer flights have less delays than mega airports with many flights but also a lot of staff?
- 8) In my Looker Studio Report, currently to look up info on specific airports you must know the airport code, I should create a key/list detailing all airport names and their corresponding codes and add the list to a new page of the report.
- 9) Look for delay abnormalities, are there specific airports that are just the absolute worst? Are their routes frequently flown with almost no delays/that often leave early?
- 10) Consider extending this analysis to a global dataset instead of one focused solely on the U.S.